INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — Cryptarithmetic & Verbal Arithmetic

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — Cryptarithmetic & Verbal Arithmetic

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/logical/cryptarithmetic-questions/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 02_Logical_Reasoning_Ability
Total Verified Questions Extracted: 12
Quizzes Covered: quiz-1, quiz-2, quiz-3

📋 Section 1: Rules to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- Each letter represents a distinct digit from 0 to 9.
- The leftmost (leading) letter of any number can NEVER be 0.
- In addition of two n-digit numbers resulting in an (n+1)-digit number, the carry digit is ALWAYS 1.
- Number Series - Coding and Number Series - Letter and Symbol Series - Logical Sequence of Words - Analogy and Classification Pattern - Statements and Conclusions - Statements and Assumptions - Data Sufficiency - Visual Reasoning - Cube and Cuboid

⚡ Section 2: How to Solve Quickly

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🚀 Section 3: Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- In SEND + MORE = MONEY, M is guaranteed to be 1, and S is either 8 or 9.
- Identify unit digit constraints first: e.g., D + E = Y or D + E = 10 + Y.
- Number Series - Coding and Number Series - Letter and Symbol Series - Logical Sequence of Words - Analogy and Classification Pattern - Statements and Conclusions - Statements and Assumptions - Data Sufficiency - Visual Reasoning - Cube and Cuboid

📚 Section 4: Practice Questions, Options & Step-by-Step Solutions

Contains **12** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
If KANSAS + OHIO = OREGON ? Then find the value of G+R+O+S+S?

(Given Value of O = 5)

K A N S A S

```
O H I O
```

O R E G O N

Options:
- **(A)** `11`
- **(B)** `20`
- **(C)** `12`
- **(D)** `18`
- **(E)** `10`

Correct Answer: (A) `11`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
KANSAS+OHIO=OREGON

K A N S A S

    O H I O

-----------

O R E G O N

given the value of O=5

Clearly,

    K + (nothing) is not equal to K

    K + (nothing) = O

    Thus, 1 carry must be coming from previous step

    K + 1 = O (O is given as 5)

    K = O - 1 = 5 - 1 = 4

Similarly,

    A + (nothing) = R

    This means 1 carry is coming from previous step

    Also, this is generating carry to next step

    Thus, A = 9

    R = A + 1(carry) = 0

rewriting the whole problem again
```

```
4 9 N S 9 S

    5 H I 5

----------

5 0 E G 5 N

If we look at the 10's digit, we find that 9 + I = 5. This is only possible when -

    Case 1 :I = 6 (no carry from previous step)

    9 + 6 = 5 (1 carry to next step)

    Case 2 : I = 5 (1 carry from previous step)

    9 + 5 + 1(carry) = 5 (1 carry to next step)

    5 value is already taken by O so not possible thus

    I = 6

Replace I = 6 and rewriting -

4 9 N S 9 S

    5 H 6 5

----------

5 0 E G 5 N

Step at units digit S + 5 = N

    This generates no carry to next step as proved above

    So S + 5 &lt; 10
```

```
    To make this possible

    Values of S ={0, 1, 2, 3, 4}

    S can't be 0 or 4 as these values are taken by R and K

    S can't be 1 as S + 5 = N

    If S is 1 then N = 6

    I is already 6 so, S can't be 1 also

    Similarly, S can't be 4

    N will become 9 and A is already 9

Let us do hit and trial with 2 and 3. Put S = 2 obviously N = 7. Let's rewrite the problem -

4 9 7 2 9 2

    5 H 6 5

-----------

5 0 E G 5 7

Now, 7 + 5 = E = 2 (1 carry to next step)

    This, is not possible As S already 2

    Thus, there must be 1 carry from previous step

7 + 5 + 1(carry) = E =&gt; E = 3

    Till now, R = 0, S = 2, K = 4, O = 5, I = 6, N = 7, A = 9
```

```
    Values pending = {1, 3, 8}

From hundreds place -

    2 + H  + 1 (carry) = G

    From the above pending values, only one case is possible when

    H = 8, G = 1

Final solution will look like -

4 9 7 2 9 2

    5 8 6 5

-----------

5 0 3 1 5 7

KANSAS=497292

OHIO=5865

OREGON=503157

G+R+O+S+S=1+0+5+2+2=10
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:

if IS+THIS=HERE then value of numeric value of T*E*+*I*R*H-S

T H I S

I S

H E R E

Options:

- **(A)** `221`
- **(B)** `178`
- **(C)** `157`
- **(D)** `149`

Correct Answer: (C) `157`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
T H I S

    I S

-------

H E R E

    T + (nothing) is not equal to H

    Thus, there must be 1 carry from previous step

    Similarly, H + (nothing) is not equal to H, thus 1 carry from previous step

    Also, H + 1 (carry) &gt;= 10 (It is generating 1 carry to next step)

This is only possible when

    H = 9

    When, H = 9, E = H + 1 = 0

Rewriting the whole problem

T 9 I S

    I S

-------

9 0 R 0

    Clearly, T = 8

    As T + 1 (carry) = 9

Also, S + S = 0
```

```
    This is only possible when S = 5

Rewriting the whole problem -

8 9 I 5

    I 5

-------

9 0 R 0

Now, I + I + 1(carry) = R is generating a carry, thus

    I = {5, 6, 7, 8, 9}

    Values : 5, 8, 9 are already taken

    Let's try I = 7

    7 + 7 + 1 = 5

    5 value is already taken

    Thus, 6 would be the value

    6 + 6 + 1 = 3

    No conflict problem is solved -

8 9 6 5

    6 5

-------

9 0 3 0
```

```
8\times 0+ 6\times 3\times 9 - 5 = 157
```

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:
GO+TO=OUT, O+U+T=?

Options:
- **(A)** `10`
- **(B)** `8`
- **(C)** `5`
- **(D)** `3`

Correct Answer: (A) `10`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
&amp;nbsp;

  T O

+ G O

-----

O U T

The above 0 is O

Step 1

    The value of O is clearly 1 , as it is a carry

    Since, T + G is generating O is carry so value of O is 1.

Now, after substituting the values -

  T 1

+ G 1

-----

1 U T

Step 2

    Value of T = 1 + 1

    Thus, value of T = 2

  2 1

+ G 1
```

```
-----

1 U 2

Step 3

    Now, 2 + G &gt; 10 (as its resulting a carry 1 on next)

    Now, possible values of G to get 1 carry at next step is - {G = 8 or 9}

If G is 9 then -

    U = 2 + 9 = 11

    So value of U becomes 1 and 1 goes to carry

    Now, value of O is already 1 so U value can not be 1 also.

Now, obv the value of G has to be 8, lets verify this also

    U = 2 + 8 = 10

    So value of U becomes 0, there is no conflict.

So final values are -

T = 2, O = 1, G = 8, U = 0

    2 1

+   8 1

-----------

  1 0 2

Now O + U + T = 1 + 0 + 2 = 3
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:
TEE + LET= ALL where E=5 find A+L+L

T 5 5

- L 5 T

A L L

Options:
- **(A)** `17`
- **(B)** `9`
- **(C)** `11`
- **(D)** `12`

Correct Answer: (C) `11`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
After replacing the value E = 5

  T 5 5

+ L 5 T

------

  A L L

    In the above example

    5 + 5 = L

    L can either be 0 or 1 (1 carry from previous step)

If L = 0

    Then, 5 + T must also be 0

    This means, T must also be 5 which is not possible

Then L = 1

    5 + T = 1( 11 as 1 carry to next step)

    So T has to be 6

    T = 6

Rewriting the problem -

  6 5 5

+ 2 5 6

------
```

```
  A 1 1


Thus, A = 6 + 2 + 1(carry) = 9


A + L + L = 9 + 1 + 1  = 11
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:
USA+USSR=PEACE FIND P+E+A+C+E

Options:
- **(A)** `10`
- **(B)** `14`
- **(C)** `25`
- **(D)** `12`

Correct Answer: (A) `10`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
U S A

+ U S S R

-----------

P E A C E

Step 1

    Clearly, P = 1, U = 9, E = 0 (1 carry from previous stage)

    Let's replace the values here

    0 S A

+ 9 S S R

-----------

1 0 A C 0

Step 2

    A + R = 0

    This is possible if, A = 5, R = 5, but, both can't take same values

    So its possible with (8,2), (7,3), (6,4), (4,6), (3,7), (2,8)

    Let's do hit and trial and take (2,8) and replace the new values.

      1

    0 S 2

+ 9 S S 8
```

```
----------

1 0 2 C 0

Step 3

    Now, assuming S value to be 3

    We have S + S + 1 (carry) = C

    3 + 3 + 1 =&gt; C = 7 (unoccupied)

    All the values are found out we just need to verify

      1

    0 3 2

+ 9 3 3 8

-----------

1 0 2 7 0

Step 4

    Values, are replaced and all the operations work just fine

Value of P + E + A + C + E = 1 + 0 + 2 + 7 + 0 = 10
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:

O N E

O N E

O N E

- O N E

T E N

Find the value of T+E+N ?

Options:

- **(A)** `17`
- **(B)** `23`
- **(C)** `19`
- **(D)** `25`

Correct Answer: (A) `17`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
  O N E

        O N E

        O N E

    +   O N E

      ----------

        T E N
```

4 x O = T

4 x N = E

4 x E = N

STEP 1:

Assume E = 1, then N = 4

Then 4N must give 1 in the units place. 4N = 4(4) = 16 which is 6 in the units place

Therefore E is not equal to 1

Assume E = 2, Then N = 4 x 2 = 8

Then 4N must give 2 in the units place. 4N = 4(8) = 32 which is 2 in the units place

Thus E = 2 and N = 8, with a carry of 3

Step 2

Let O = 1, then T = 4 x 1 = 4 + 3(carry) = 7

We cannot take O other than 1 as it gives a carry

```
Thus,

    3

     1 8 2

     1 8 2

     1 8 2

 +   1 8 2

    --------

     7 2 8

Thus T = 7 , E = 2, N = 8

T + E + N = 7 + 2 + 8 = 17
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:

EAT+EAT+EAT=BEET if T=0 then what will the value of TEE+TEE

E A T

E A T

E A T

B E E T

Options:

- **(A)** `088`
- **(B)** `077`
- **(C)** `066`
- **(D)** `055`

Correct Answer: (A) `088`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Putting T = 0 as given

  E A 0

  E A 0

  E A 0

--------

B E E 0

    As T=0, no carry for A+A+A = (3A).

Since, the step above can generate following possible carries C = {0, 1, 2}

  C

  E A 0

  E A 0

  E A 0

--------

B E E 0

    There will be no carry generate from units place to tens place as all values are 0

Possible values of A and E can be calculated by

    3A= E (No Carry to next step) - Eq1

    3A = 10 +E (1 Carry to next step) - Eq2

    3A = 20 + E (2 Carry to next step) - Eq3
```

```
Here,

    Largest carry generated by addition of three one digit number is 27(9+9+9).

    Hence value of E is less than 7 for Eq 3

    For Eq 3

Assume value of E is 7.

    Therefore value of A=9

    Now, C + E + E + E = BE.

    (2) + (7) + (7) + (7) = 23.

    This contradicts are resultant should also be 7, while its 3

We tried A = 9, for which E was 7 and resulted in contradiction.

Let us now try, A = 8

    If A = 8

    Then, 3A = E + (carry)

    Putting A = 8, E = 4 (carry = 2)

Trying the same again,

    Now, C + E + E + E = BE.

    (2) + (4) + (4) + (4) = 14.

    Hence value satisfied with our prediction.

    Thus, E=4 A=8 and B=1
```

```
Putting in equation - 2




  4 8 0

  4 8 0

  4 8 0

--------

1 4 4 0

Finally, T E E + T E E = 044 + 044 = 088
```

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:
If CROSS + ROADS = DANGER then D+A+N+G+E+R=?

C R O S S

R O A D S

D A N G E R

Options:
- **(A)** `31`
- **(B)** `21`
- **(C)** `11`
- **(D)** `16`

Correct Answer: (A) `31`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
C R O S S

  R O A D S

----------

D A N G E R

    Clearly, D = 1

    C + R can at max generate 1 as carry

Now, Let us look at the step S + S = R

    Clearly, R would be even, as sum of S + S will always be even

    So, possible values for R = {0, 2, 4, 6, 8}

    For R = 0

    S needs to be 0

    Both S and R can't be 0 thus, not possible

    For R = 2

    S needs to be 1

    D is already 1 so not possible

    R = 4

    S = 2

    Possible (Case 1)

    R = 6
```

```
    S = 3

    Possible (Case 2)

    R = 8

    S = 4

    Possible

Let us start hit and trial with S = 2 and R = 4

Putting all these values and rewriting -

Assuming C1, C2, C3 are carries, units, tens and hundreds will not have carry obv.

C1 C2 C3

    C  4  O  2  2

    4  O  A  1  2

-----------------

 1  A  N  G  E  4

    Clearly, E = 3 and C1 = 1

    Now, C2 + C + 4 = A (1 carry to next step)

    C2 + C + 4 &gt;= 10

    Thus, C2 + C &gt;= 6

    Possible Values for C2 = {0,1}

    C = {9, 8, 7, 6}
```

```
Trying Hit and Trial with

    C = 9, C2 = 1

    C2 + C + 4 = A

    1 + 9 + 4 =&gt; A = 4 (1 carry)

    Conflict as R already 4

    C = 9, C2 = 0

    After, substituting A = 3

    Conflict E = 3

    Let us try C = 8, C2 = 1

    After, substituting, A = 3

    Conflict E = 3

    C = 8, C2 = 0

    A = 2

    Conflict, S = 2

Similarly we can try all other values of C = 7, 6 with C1 = 1,0 all will have conflicts

Going back and trying Case 2, S = 3 and R = 6

C1 C2 C3

    C  6  O  3  3
```

```
    6  O  A  1  3

-----------------

 1  A  N  G  4  6

    Clearly, E = 4 and C1 = 1

    Now, C2 + C + 6 = A (1 carry to next step)

    C2 + C + 6 &gt;= 10

    Thus, C2 + C &gt;= 4

    Possible Values for C2 = {0,1}

    C = {9, 8, 7, 5} (4, 6 values already taken)

Taking C = 9 and C2 = 0

    C2 + C + 6 = A

    0 + 9 + 6 =&gt; A = 5

Rewriting all values -

 1  0 C3

    9  6  O  3  3

    6  O  A  1  3

-----------------

 1  5  N  G  4  6

    Clearly, Step 6 + O = N is not generating any carry.
```

```
    Thus, 6 + O &lt;=9

    O &lt;=3, Possible values are O = {3, 2, 1, 0}

Let us try with O = 3

    C3 + 6 + 3 = N

    If C3 = 1

    1 + 6 + 3 = N

    N = 0 (1 carry, not possible as C2 was found to be 0)

    If C2 = 0

    N = 9

    Not Possible as C is already 9

Let us try with O = 2

    C3 + 6 + 2 = N

     If C3 = 1

    N = 9, not possible as C = 9

    If C3 = 0

    N = 8

    Possible

Rewriting all values -

 1  0  0
```

```
    9  6  2  3  3

    6  2  A  1  3

------------------

 1  5  8  G  4  6

    Values taken D = 1, O = 2, S = 3, E = 4, R = 6, N = 8,  C = 9

    Left values = {0, 5, 7}

    A = 5, G = 7, Clearly satisfies the conditions.

CROSS= 96233

ROADS=62513

DANGER=158746

Ans 31
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:
Fine+Nine= Wives then W+I+V+E+S=?

F I N E

N I N E

W I V E S

Options:
- **(A)** `22`
- **(B)** `15`
- **(C)** `33`
- **(D)** `37`

Correct Answer: (A) `22`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
F I V E

  N I N E

---------

W I V E S

    Clearly, W = 1, as F + N = WI (2 digit number)

Now,

    N + N = E =&gt; 2N = E

    E + E = S =&gt; (2N) + (2N) = S

    4N = S

Possible digits to satisfy this between  {1 to 9} would be -

    N = 1, E = 2, S = 4

    or N = 2, E = 4, S = 8

But, W already is 1. So, N can not be 1

Thus, N = 2, E = 4, S = 8 are only options....

Rewriting the problem -

  F I 2 4

  2 I 2 4

---------

1 I V 4 8
```

```
    F + 2 + carry(0/1) &gt;=10 (as 1 carry to next step)

    To do this possible values of F are = {7, 8, 9}

    8 is already taken by S, So, F = {7,9}

If we take F = 7

    F + 2 + Carry(0/1) = I

    9 + Carry(0/1) = I

    9 + 0 (carry) = I (Case 1)

    This is not possible as no carry to next step

    9 + 1 (carry) = I (case 2)

    I = 0 not possible

    As step I + I = V should generate carry to next step i.e. Case 2

We must have done something wrong what?

There is another case that we never noticed -

    What if the below equations were never valid as they were generating carries

    N + N = E =&gt; 2N = E

    E + E = S =&gt; (2N) + (2N) = 6

    4N = S

Let us check

    What if E + E at units digit was generating a carry to next step
```

```
    Possible values to do this for E are = {5, 6, 7, 8, 9}

Let us try with E = 5

    E + E = S

    =&gt; S = 0

    Also, N + N + 1(carry) = E

    N + N + 1 = 5

    Possible values of N to do this are N = {7, 2}

Let us try N = 7 and other values and rewrite the equation -

  F I 7 5

  7 I 7 5

---------

1 I V 5 0

    F + 7 = 1I

    =&gt; F + 7 + carry(0/1) &gt;= 10

Let us assume the case where carry 1 is coming from previous step (Note 1)

    =&gt; F + 7 + 1 &gt;= 10

    Equation is F + 7 + 1 = I

    Possible values for F are ={2, 3, 4, 6, 8, 9}
```

F = 2 not possible as it will result I = 0, S is already 0

F = 3 not possible as it will result I = 1, W already 1

F = 4 is possible as I will be 2

But, step I + I + 1(Carry) = V will not generate carry as

2 + 2 = 4 and carry is needed for (Note 1)

F = 6, is possible and I = 4

But, again I + I + 1(Carry) = V will not generate carry

F = 8, is possible and I = 6

I + I + 1(Carry) = V will generate carry

6 + 6 + 1 = V = > V = 3

Putting all of these values in equation

8 6 7 5

7 6 7 5

---

1 6 3 5 0

Thus, W + I + V + E + S = 15

Many websites give this solution which is incorrect

w=1

n=2

```
e=4

s=8

i=3

v=6

summation of wives =22.
```

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:

If Ever + Since = Darwin then D + A + R + W + I + N is ?

```
 E V E R
```

S I N C E

D A R W I N

Options:

- **(A)** `23`
- **(B)** `41`
- **(C)** `34`
- **(D)** `16`

Correct Answer: (A) `23`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
E V E R

  9 I N C E

-----------

1 0 R W I N
```

Clearly, D = 1

Clearly S = 9

As one carry must have been from previous step

A = 0

2. Then, we looks at E and I which results in R, as

9 is already occupied by S so E could be 8 and I

could be 7 and R will be 5 with carry 1.

3. Then we place the values of R=5 and E=8

resulting in N=3 with carry 1.

4. In ten’s digits – E=8 and and value of C could

be 3,4,6 (As all others are occupied) and to make I=7 (previously assigned in step 2) E must

be added by 9 which is already occupied by S.

So this hit ends here.

Now start again with reduced value of E or I,

repeat above steps until you get a correct

```
answer or a dead end.

5. Repeating above steps one time comes with

E=5 and I=7 which results in R=3 carry=1.

6. Placing R=3 and E=5 in unit digits and their

sum gives 8.

7. in ten’s digits place E=5 now you can add 2 to

make I=7 hence C=2.

8. in 100’s digits you have N=8 (from step 6)

and only digits that are left now are 4 and 6 hence. So placing V=4 you will get W=2 which

is not true because it is already assigned to C.

So we are left with only V=6 as the ultimate

choice and this leads the result in to W=4.

Hence this way we got the answer

    5 6 5 3

  9 7 8 2 5

-----------

1 0 3 4 7 8

Hence D+A+R+W+I+N = 1+0+3+4+7+8 =23
```

Question 11 (PrepInsta quiz-1 - Q11)

Question Stem:
If (HE)^H=SHE, where the alphabets take the values from (0-9) & all the alphabets are single digit then find the value of (S+H+E)?

Options:
- **(A)** `14`
- **(B)** `19`
- **(C)** `13`
- **(D)** `15`

Correct Answer: (C) `13`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Since, SHE is a 3 digit number.

    Range of SHE is 100 - 999.

    In fact. true range would be 102 - 987

    Numbers can repeated in SHE

H can be anything between = {2, 3, 4, 5, .......}

Let us try hit and trial with 2

The equation will change as :

    (2E)^2 = S2E

    (HE)^H must be in the range 21 to 29

    so that result must be less then 999

    Do hit and trial and you will find answer is :

    (25)^2=625.

so S+H+E=6+2+5=13 ANS
```

Question 12 (PrepInsta quiz-1 - Q12)

Question Stem:
WORLD+TRADE=CENTER value of C+E+N+T+E+R

W O R L D

T R A D E

C E N T E R

Options:
- **(A)** 27
- **(B)** 24
- **(C)** 12
- **(D)** 19

Correct Answer: (C) 12

PrepInsta Detailed Explanation & Mathematical Derivation:

```
WORLD+TRADE=CENTER

  W O R L D

  T R A D E

-----------

C E N T E R

ANSWER CENTER

so,start from last c the value is 1 always.

Next W+T=E right.

By adding numbers we should get 1 as a carry (which is C value).

so,take W as 5 and T as 7 and add now which is 12.

NO TWO ALPHABETS SHOULD HAVE SAME VALUES ALWAYS.

Do it accordingly.

  5 3 6 8 4

  7 6 0 4 2

-----------

1 2 9 7 2 6

53684+ 76042=129726
```